Fundamentos Físicos I

Soluciones Problemas Tema 4

1) a) $\Phi = 0$

b) $\vec{F} = 0$; $\vec{M} = -qlE\sin\alpha\,\vec{k}$

2) $q' = -\frac{\epsilon_r - 1}{\epsilon_r} q$

3) a) $Q_{fin}/Q_{ini} = 1$; $E_{fin}/E_{ini} = 1$; $\phi_{fin}/\phi_{ini} = 7$; $C_{fin}/C_{ini} = 1/7$; $U_{fin}/U_{ini} = 7$

b) $Q_{fin}/Q_{ini} = 1$; $E_{fin}/E_{ini} = 1/7$; $\phi_{fin}/\phi_{ini} = 1/7$; $C_{fin}/C_{ini} = 7$; $U_{fin}/U_{ini} = 1/7$

c) $Q_{fin}/Q_{ini} = 1/7$; $E_{fin}/E_{ini} = 1/7$; $\phi_{fin}/\phi_{ini} = 1$; $C_{fin}/C_{ini} = 1/7$; $U_{fin}/U_{ini} = 1/7$

d) $Q_{fin}/Q_{ini} = 7$; $E_{fin}/E_{ini} = 1$; $\phi_{fin}/\phi_{ini} = 1$; $C_{fin}/C_{ini} = 7$; $U_{fin}/U_{ini} = 7$

4) a) $\vec{D}(r_1 = 1.2\,cm) = -2.76 \times 10^{-6}\,\vec{u}_r$ C/m^2; $\vec{E}(r_1 = 1.2\,cm) = -7.81 \times 10^4\,\vec{u}_r$ V/m; $\vec{P}(r_1 = 1.2\,cm) = -2.07 \times 10^{-6}\,\vec{u}_r$ C/m^2

b) $\vec{D}(r_2 = 2.2\,cm) = 0$ C/m^2; $\vec{E}(r_2 = 2.2\,cm) = 0$ V/m; $\vec{P}(r_2 = 2.2\,cm) = 0$ C/m^2

c) $q' = -3.75 \times 10^{-9}$ C

5) a) $q_{1,izq} = +C\Delta\phi$; $q_{1,der} = -C\Delta\phi$; $q_{2,izq} = +C\Delta\phi$; $q_{2,der} = -C\Delta\phi$

b) $\vec{E}_1 = \Delta\phi/d\,\vec{i}$; $\vec{E}_2 = \Delta\phi/d\,\vec{i}$

c) $q_{1,izq} = +C\Delta\phi$; $q_{1,der} = -C\Delta\phi$; $q_{2,izq} = +\epsilon_r C\Delta\phi$; $q_{2,der} = -\epsilon_r C\Delta\phi$

d) $\vec{E}_1 = \Delta\phi/d\,\vec{i}$; $\vec{E}_2 = \Delta\phi/d\,\vec{i}$

6) a) $C_0 = 8.85$ pF

b) $Q_0 = 8.85 \times 10^{-10}$ C

c) $E_{hueco} = 10^4$ V/m; $E_{diel} = 1.43 \times 10^3$ V/m

d) $\Delta\phi = 57$ V

e) $C_0 = 15.5$ pF

f) $D_{hueco} = 8.85 \times 10^{-8}$ C/m^2; $D_{diel} = 8.85 \times 10^{-8}$ C/m^2; $P_{hueco} = 0$ C/m^2; $P_{diel} = 7.59 \times 10^{-8}$ C/m^2

7) $U_{ini} = \frac{1}{2}(C_1\phi_1^2 + C_2\phi_2^2)$

$U_{fin} = \frac{1}{2}\frac{(C_1\phi_1 + C_2\phi_2)^2}{C_1 + C_2}$;

$U_{fin} < U_{ini}$

8) a) $q_1^{fin} = 0$; $q_2^{fin} = q_1 + q_2$

b) $U_{fin} - U_{ini} = -\frac{q_1^2(R_2 - R_1)}{8\pi\epsilon_0 R_1 R_2}$

9) a) $U = \frac{1}{2}CV_0^2 = \frac{1}{2}\frac{\varepsilon_0 S}{d}V_0^2$

b) $\Delta U = \frac{\varepsilon_0 S}{10d}V_0^2$

c) $\Delta Q = \frac{\varepsilon_0 S}{5d} V_0$

d) $\Delta U = \frac{\varepsilon_0 S}{4d} V_0^2$

10) a) $\sigma'(R) = \frac{q}{12\pi R^2}$; $\sigma'(2R) = -\frac{q}{48\pi R^2}$

b) $V(R) = 0$.

c) $u(r < R) = 0$; $u(R < r < 2R) = \frac{q^2}{144\pi^2 \varepsilon_0 r^4}$; $u(r > 2R) = \frac{q^2}{288\pi^2 \varepsilon_0 r^4}$

b) $U_{fin} - U_{ini} = 0$

11) a) $Q_1 = \frac{2\epsilon_r C_0 \phi_0}{1+\epsilon_r}$; $Q_2 = \frac{2C_0\phi_0}{1+\epsilon_r}$

b) $\phi_1 = \phi_2 = \frac{2\phi_0}{1+\epsilon_r}$

c) $U_{fin} - U_{ini} = -C_0\phi_0^2 \frac{\epsilon_r - 1}{\epsilon_r + 1}$

d) $D_1 = Q_1/A$; $D_2 = Q_2/A$